03: Inference for SLR

Stat 230 - Spring 2026

```r
library(tidyverse)
library(broom)
```

First, skim working in R for Stat230. Then, follow these directions:

1. Either open your local RStudio program or navigate to https://maize.mathcs.carleton.edu in your browser and log in.
2. Go to file –> New file –> Quarto document … –> "Create empty document". This should open a new .qmd file in your Rstudio session. Delete everything in this file so that you have a totally blank document.
3. Next, we need to copy this template into your new quarto file.

- come back to this page and click the "code" button at the top of the page.
- Click the "copy" symbol in the upper right corner of the popup file
- Navigate back to your empty .qmd file and "paste" the text there

4. You should now be able to see this document in your own session, and you can run code and edit it as you need to! Make sure to save it in a "Stat230" folder

1 Part 2: In R[1]

1.1 Load the data

Load data into your session (it should show up in the "Environment" pane)

[1] Adapted from *Stat 2: Modeling with Regression and ANOVA*

```r
LeafWidth <- read_csv("https://aluby.github.io/stat230/data/LeafWidth.csv")
```

"Print" the data:

```r
LeafWidth
```

```
# A tibble: 252 x 5
   Width Length LWRatio  Area  Year
   <dbl>  <dbl>   <dbl> <dbl> <dbl>
 1 0.658   85.1   129.   70.3  1974
 2 0.921   53.2    57.7  49.0  1918
 3 0.921   59.9    65    57.7  1993
 4 0.921   63.7    69.1  37.4  1946
 5 0.921   64.2    69.7  62.6  1956
 6 0.921   87.0    94.4  67.5  1964
 7 1.05    61.3    58.2  63.9  1956
 8 1.05    64.1    60.9  54.0  1980
 9 1.18    41.2    34.8  40.9  1999
10 1.18    41.7    35.2  34.6  1977
# i 242 more rows
```

1.2 Exploratory Data Analysis

First, make a scatterplot of `Width` (response variable) against `Year` (explanatory variable). Overlay the linear regression line. Describe your findings.

1.3 Linear Regression Model

Find the linear regression equation using `lm()`. Use `tidy()` to print the coefficient table of the regression model. Make sure you get the same answer as the paper printout!

1.4 CI for Slope

Calculate a 95% confidence interval for the slope of the regression line using `confint`.